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Project: Capstone 2 - JavaScript Library Debug

## List of Critical errors with impact assessment

1. **Missing variable declarations** (books, MAX_BOOKS_PER_MEMBER ) – Causes ReferenceError in strict mode
2. **Missing super() in DigitalBook constructor** – DigitalBook objects cannot be instantiated.
3. **Assignment instead of comparison in findMemberById()** – Always returns the wrong member, breaking all member-related operations.
4. **Assignment instead of comparison in handleFilterChange()** – Category filtering fails.
5. **Infinite loop in** processReturnQueue() – Causes the application/browser to freeze
6. **Missing base case in searchBooksByCategory()** – Causes a stack overflow and crashes category searches.
7. **Assignment instead of comparison in canBorrow()** – Borrowing limits are never enforced, violating core business rules.
8. **No book availability validation in checkOut()** – Allows unlimited borrowing.
9. **importLibraryData() lacks error handling and object reconstruction** – Imported data becomes unusable.
10. **saveToLocalStorage() lacks JSON.stringify()** – Data is not saved correctly
11. **loadFromLocalStorage() lacks null checks and JSON.parse()** – Saved library data cannot be loaded.
12. **Incorrect script loading order** (ui.js before library.js) – Application fails to start.
13. **initializeUI() executes before DOMContentLoaded** – UI initialization fails.
14. **calculateTotalLateFees() references a non-existent property** (memberRecord.overdueBooks) – Throws a TypeError.
15. **var members used instead of let/const** – Function scoping and hoisting risk unintended modification of global member data, compromising application state.
16. missingng joinDate and getMembershipDuration() in Member – Calling getMembershipDuration() throws a TypeError.

## List of major issues categorized by JavaScript concept

1. **PremiumMember constructor lacks premium properties** (maxBooks, discountRate, priorityHolds) – Premium features unavailable.
2. **DigitalBook.download() does not record memberId or checkoutDate** – Downloaded books cannot be tracked for overdue checks.

3. **Member.canBorrow() lacks duplicate-book validation** – Duplicate borrowing allowed
4. **borrowBook() lacks input validation and error handling** – Invalid input may cause runtime errors.
5. **handleBorrowSubmit() does not call event.preventDefault()** – Form reload interrupts borrowing.
6. **Incorrect selector in filterDropdown** (document.querySelector("filter-category")) – Element is never found.
7. **Incorrect event type in filterDropdown** – Category filtering fails.

## List of missing features and implementations required

1. **Overdue book tracking** – Overdue tracking – Returns overdue books with overdue days and fines.
2. **Book return workflow** – Supports multi-book returns and prompts for overdue fee confirmation only when a book is overdue.
3. **Book editing** – Supports editing, ISBN validation, copy updates and Physical-Digital conversion.
4. **Member editing** – Member editing – Supports editing, email validation, and premium downgrade rules.
5. **Server persistence** – Saves library data through an Express POST /save endpoint.

## Proposed systematic fix strategy

**Phase 1 — Variable Declarations & Script Order**
Replace all implicit globals with let/const. Correct the script loading order in index.html (library.js before ui.js).

**Phase 2 — Class Correctness**
Add the missing super() call to DigitalBook. Add missing properties. correct comparison operators and implement validations.

**Phase 3 — Loop & Algorithm Improvements**
Refactor loops, fix recursion and infinite loops, and use rest/spread operators.

**Phase 4 — Error Handling**
Add try...catch blocks, input validation, and null checks.

**Phase 5 — DOM, Events & Storage**
Fix DOM events, selectors, storage, and rendering. Use DocumentFragment for efficient batch rendering.

**Phase 6 — New Feature Implementation**
Implement overdue tracking, book return workflow, book and member editing, server persistence, and digital book support.